Mathematics-II (DMA025A)

Mathematics II (DMA025A)

Course PPT

Unit-IV: Complex Variable-Differentiation

Department of Mathematics

JECRC University Jaipur

DMA025A	Mathematics-II	3+1+0 = 4 Credits

Course Contents:

UNIT I: Matrices:
Linear Systems of Equations, Linear Independence, Rank of a Matrix; Determinant, Inverse of a matrix, rank-nullity theorem, System of linear equations, Symmetric, skew-symmetric and orthogonal matrices, Determinants, Eigen values and eigenvectors, Orthogonal transformation, Diagonalization of matrices, Cayley-Hamilton Theorem.

UNIT II: First order ordinary differential equations:
Exact, linear and Bernoulli's equations. Equations not of first degree: equations solvable for p, equations solvable for y, equations solvable for x and Clairaut's type.

UNIT III: Ordinary differential equations of higher orders:
Second order linear differential equations with variable coefficients: Euler-Cauchy equations, solution by variation of parameters; Power series solutions: Legendre's equations and Legendre polynomials, Frobenius method, Bessel's equation and Bessel's functions of the first kind and their properties.

UNIT IV: Complex Variable–Differentiation:

Differentiation, Cauchy-Riemann equations, analytic functions, harmonic functions, finding harmonic conjugate, elementary analytic functions(exponential, trigonometric, logarithm) and their properties, Conformal mappings, Mobius transformations and their properties.

UNIT V: Complex Variable–Integration:
Contour integrals, Cauchy-Goursat theorem (without proof), Cauchy Integral formula (without proof), Liouville's theorem and Maximum-Modulus theorem (without proof); Taylor's series, zeros of analytic functions, singularities, Laurent's series; Residues, Cauchy Residue theorem (without proof), Evaluation of definite integral involving sine and cosine, Evaluation of certain improper integrals using the Bromwich contour.

Text/Reference Books:

- AICTE's Prescribed Textbook: Mathematics-II (Calculus, Ordinary Differential Equations and Complex Variable), Khanna Book Publishing Co.
- Reena Garg, Engineering Mathematics, Khanna Book Publishing Company, 2022.
- Reena Garg, Advanced Engineering Mathematics, Khanna Book Publishing Company, 2021.
- Erwin Kreyszig, Advanced Engineering Mathematics, 10th Edition, John Wiley & Sons, 2006.
- Veerarajan T., Engineering Mathematics for first year, Tata McGraw-Hill, New Delhi, 2008.
- B. S. Grewal, Higher Engineering Mathematics, Khanna Publishers, 36th Edition, 2010.

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Course Outcome (CO):

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Upon successful completion of this course, the student will be able to:

- CO1: The students will learn the essential tool of matrices and linear algebra in a comprehensive manner.
- CO2: Understand the linear ordinary differential equations along with their applications.
- CO3: Will able to solve problem involving ordinary differential equations of higher orders.
- CO4: The students will learn the Complex Variable – Differentiation.
- CO5: The students will learn the Complex Variable–Integration.

Mapping of COs with Pos

MAPPING COURSE OUTCOMES LEADING TO THE ACHIEVEMENT OF PROGRAM OUTCOMES AND PROGRAM SPECIFIC OUTCOMES:

Course Outcome	Program Outcome												Program Specific Outcome		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	M	L			M		M					M		M	
CO2	M	M		L	M		M			M	M	M	M		
CO3		L			M					L		M	M	M	
CO4	M	M		M	L		M					M		M	
CO5	M			M	M		H					M	M	M	

H=Highly Related ;M=Medium L=Low

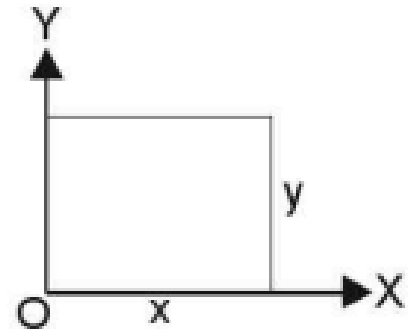
Contents

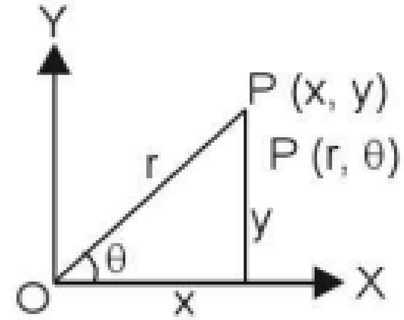
- Complex variable and function.
- Limit, continuity of a complex function, Differentiation.
- Cauchy-Riemann equations (Cartesian and Polar form).
- Analytic functions, harmonic functions.
- Finding harmonic conjugate, Milne Thomson's Method.
- Elementary analytic functions (exponential, trigonometric, logarithm) and their properties.
- Conformal mappings, and Mobius transformations and their properties

COMPLEX VARIABLE

$x + iy$ is a complex variable and it is denoted by z.

(1) $z = x + iy$. where $i = \sqrt{-1}$ (Cartesian form)





(2) $z = r(\cos\theta + i\sin\theta)$(Polar form)

(3) $z = re^{i\theta}$ (Exponential form)

FUNCTIONS OF A COMPLEX VARIABLE

$f(z)$ is a function of a complex variable z and is denoted by w.

$$w = f(z)$$

$$w = u + iv$$

where u and v are the real and imaginary parts of $f(z)$.

Limit of a function

27.4 LIMIT OF A FUNCTION OF A COMPLEX VARIABLE

Let $f(z)$ be a single valued function defined at all points in some neighbourhood of point z_0. Then the limit of $f(z)$ as z approaches z_0 is w_0.

$$\lim_{z \to z_0} f(z) = w_0$$

Example. Prove that $\displaystyle\lim_{z \to 1-i} \frac{(z^2+4z+3)}{z+1} = 4-i$

Solution. $\displaystyle\lim_{z \to 1-i} \frac{z^2+4z+3}{z+1} = \lim_{z \to 1-i} \frac{(z+1)(z+3)}{(z+1)} = \lim_{z \to 1-i} (z+3) = (1-i)+3 = 4-i$ **Proved.**

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CONTINUITY

The function $f(z)$ of a complex variable z is said to be continuous at the point z_0 if for any given positive number ϵ, we can find a number δ such that $|f(z) - f(z_0)| < \epsilon$

for all points z of the domain satisfying

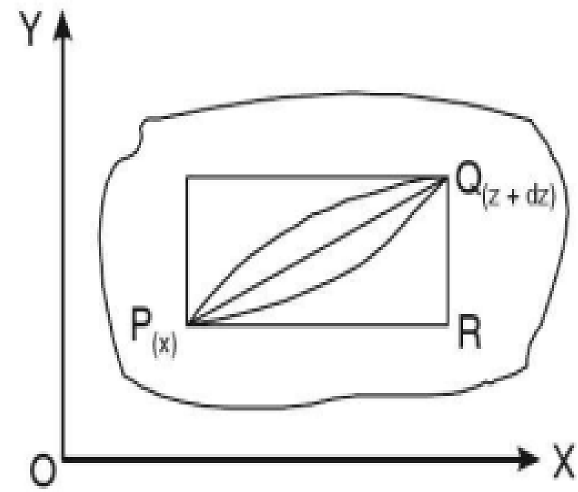
$$|z - z_0| < \delta$$

$f(z)$ is said to be continuous at $z = z_0$ if

$$\lim_{z \to z_0} f(z) = f(z_0)$$

Differentiability

DIFFERENTIABILITY



Let $f(z)$ be a single valued function of the variable z, then

$$f'(z) = \lim_{\delta z \to 0} \frac{f(z+\delta z) - f(z)}{\delta z}$$

provided that the limit exists and is independent of the path along which $\delta z \to 0$.

Let P be a fixed point and Q be a neighbouring point. The point Q may approach P along any straight line or curved path.

Example 1

Example 1. *Consider the function*

$$f(z) = 4x + y + i(-x + 4y)$$

and discuss $\dfrac{df}{dz}$.

Solution. Here, $f(z) = 4x + y + i(-x + 4y)$

$$= u + iv$$

so $$u = 4x + y$$

and $$v = -x + 4y$$

$$f(z+\delta z) = 4(x+\delta x) + (y+\delta y) - i(x+\delta x) + 4i(y+\delta y)$$

$$f(z+\delta z) - f(z) = 4(x+\delta x) + (y+\delta y) - i(x+\delta x) + 4i(y+\delta y) - 4x - y + ix - 4iy$$

$$= 4\delta x + \delta y - i\,\delta x + 4i\delta y$$

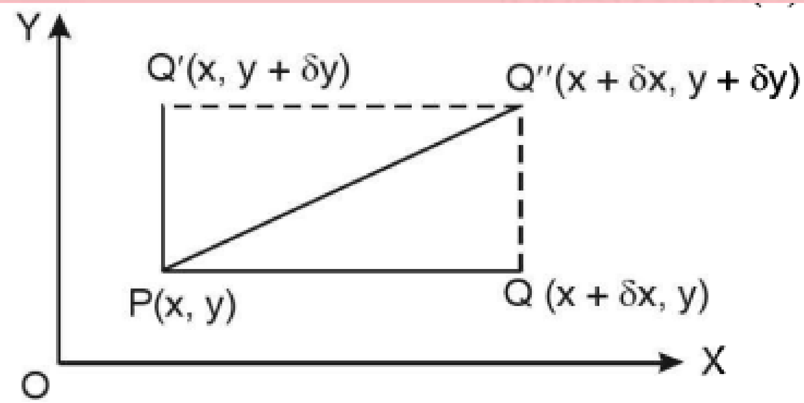
$$\frac{f(z+\delta z) - f(z)}{\delta z} = \frac{4\delta x + \delta y - i\,\delta x + 4i\delta y}{\delta x + i\delta y}$$

$$\Rightarrow \qquad \frac{\delta f}{\delta z} = \frac{4\delta x + \delta y - i\delta x + 4i\delta y}{\delta x + i\delta y} \qquad \ldots (1)$$

Contd…

(*a*) **Along real axis:** If Q is taken on the horizontal line through P (x, y) and Q then approaches P along this line, we shall have $\delta y = 0$ and $\delta z = \delta x$.

$$\frac{\delta f}{\delta z} = \frac{4\delta x - i\delta x}{\delta x} = 4 - i$$



(*b*) **Along imaginary axis:** If Q is taken on the vertical line through P and then Q approaches P along this line, we have

$$z = x + iy = 0 + iy,\ \ \delta z = i\delta y,\ \delta x = 0.$$

Putting these values in (1), we have

$$\frac{\delta f}{\delta z} = \frac{\delta y + 4i\delta y}{i\delta y} = \frac{1}{i}(1 + 4i) = 4 - i$$

(*c*) **Along a line $y = x$:** If Q is taken on a line $y = x$.

$$z = x + iy = x + ix = (1 + i)\,x$$

$$\delta z = (1 + i)\,\delta x \quad \text{and} \quad \delta y = \delta x$$

Contd…

On putting these values in (1), we have

$$\frac{\delta f}{\delta z}=\frac{4\delta x+\delta x-i\delta x+4i\delta x}{\delta x+i\delta x}=\frac{4+1-i+4i}{1+i}=\frac{5+3i}{1+i}=\frac{(5+3i)(1-i)}{(1+i)(1-i)}=4-i$$

In all the three different paths approaching Q from P, we get the same values of $\frac{\delta f}{\delta z}=4-i$.

In such a case, the function is said to be differentiable at the point z in the given region.

Example 2

Example 2. *If* $f(z) = \begin{cases} \dfrac{x^3y(y-ix)}{x^6+y^2}, & z \neq 0, \\ 0, & z = 0 \end{cases}$ *then discuss* $\dfrac{df}{dz}$ *at* $z = 0$.

Solution. If $z \to 0$ along radius vector $y = mx$

$$\lim_{z\to 0}\frac{f(z)-f(0)}{z} = \lim_{z\to 0}\left[\frac{\dfrac{x^3y(y-ix)}{x^6+y^2}-0}{x+iy}\right] = \lim_{z\to 0}\left[\frac{-ix^3y(x+iy)}{(x^6+y^2)(x+iy)}\right]$$

$$= \lim_{z\to 0}\left[\frac{-ix^3y}{x^6+y^2}\right] = \lim_{z\to 0}\left[\frac{-ix^3(mx)}{x^6+m^2x^2}\right] \qquad [\because \; y = mx\,]$$

$$= \lim_{x\to 0}\left[\frac{-imx^2}{x^4+m^2}\right] = 0$$

But along $y = x^3$

$$\lim_{z\to 0}\frac{f(z)-f(0)}{z} = \lim_{z\to 0}\left[\frac{-ix^3y}{x^6+y^2}\right] = \lim_{x\to 0}\frac{-ix^3(x^3)}{x^6+(x^3)^2} = -\frac{i}{2}$$

In different paths we get different values of $\dfrac{df}{dz}$ *i.e.* 0 and $\dfrac{-i}{2}$. In such a case, the function is not differentiable at $z = 0$.

Theorem 1

Theorem: Continuity is a necessary condition but not sufficient condition for the existence of a finite derivative.

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Example 3

Example 3. *Prove that the function $f(z) = |z|^2$ is continuous everywhere but no where differentiable except at the origin.*

Solution. Here, $f(z) = |z|^2$.

$\therefore$ But $|z| = \sqrt{(x^2+y^2)} \quad \Rightarrow \quad |z|^2 = x^2 + y^2$

Since x^2 and y^2 are polynomial so $x^2 + y^2$ is continuous everywhere, therefore, $|z|^2$ is

Now, we have $f'(z) = \lim\limits_{\delta z \to 0} \dfrac{f(z+\delta z) - f(z)}{\delta z}$

$$f'(z) = \lim_{\delta z \to 0} \frac{|z+\delta z|^2 - |z|^2}{\delta z} \qquad (z\bar{z} = |z|^2)$$

$$= \lim_{\delta z \to 0} \frac{(z+\delta z)(\bar{z}+\delta\bar{z}) - z\bar{z}}{\delta z} = \lim_{\delta z \to 0} \frac{z\bar{z} + z\delta\bar{z} + \delta z.\bar{z} + \delta z.\delta\bar{z} - z\bar{z}}{\delta z}$$

$$= \lim_{\delta z \to 0} \frac{z.\delta\bar{z} + \delta z.\bar{z} + \delta z\delta\bar{z}}{\delta z} = \lim_{\delta z \to 0}\left\{\bar{z} + \delta\bar{z} + z\frac{\delta\bar{z}}{\delta z}\right\} = \lim_{\delta z \to 0}\left\{\bar{z} + z\frac{\delta\bar{z}}{\delta z}\right\} \quad \ldots(1)$$

[Since, $\delta z \to 0$ so $\delta\bar{z} \to 0$]

Let $\delta z = r(\cos\theta + i\sin\theta)$ and $\delta\bar{z} = r(\cos\theta - i\sin\theta)$

Contd…

$$\Rightarrow \quad \frac{\delta \bar{z}}{\delta z} = \frac{\cos\theta - i\sin\theta}{\cos\theta + i\sin\theta} \qquad \Rightarrow \quad \frac{\delta \bar{z}}{\delta z} = (\cos\theta - i\sin\theta)(\cos\theta + i\sin\theta)^{-1}$$

$$\Rightarrow \quad \frac{\delta \bar{z}}{\delta z} = (\cos\theta - i\sin\theta)(\cos\theta - i\sin\theta) \quad \Rightarrow \quad \frac{\delta \bar{z}}{\delta z} = (\cos\theta - i\sin\theta)^2$$

$$\Rightarrow \quad \frac{\delta \bar{z}}{\delta z} = \cos 2\theta - i\sin 2\theta$$

Since $\frac{\delta \bar{z}}{\delta z}$ depends on θ. It means for different values of θ, $\frac{\delta \bar{z}}{\delta z}$ has different values.

It means $\frac{\delta \bar{z}}{\delta z}$ has different values for different z.

Therefore $\lim_{\delta z \to 0} \frac{\delta \bar{z}}{\delta z}$ does not tend to a unique limit when $z \neq 0$.

Thus, from (1), it follows that $f'(z)$ is not unique and hence $f(z)$ is not differentiable when $z \neq 0$.

But when $z = 0$ then $f'(z) = 0$ *i.e.*, $f'(0) = 0$ and is unique.

Hence, the function is differentiable at $z = 0$. **Ans.**

ANALYTIC FUNCTION

A function $f(z)$ is said to be **analytic** at a point z_0, if f is differentiable not only at z_0 but at every point of some neighbourhood of z_0.

A function $f(z)$ is analytic in a domain if it is **analytic** at every point of the domain.

The point at which the function is not differentiable is called a **singular point** of the function.

An analytic function is also known as "holomorphic", "regular", "monogenic".

Entire Function. A function which is analytic everywhere (for all z in the complex plane) is known as an entire function.

For Example 1. Polynomials rational functions are entire.

2. $|\bar{z}|^2$ is differentiable only at $z = 0$. So it is no where analytic.

Note: (*i*) An entire is always analytic, differentiable and continuous function. But converse is not true.

(*ii*) Analytic function is always differentiable and continuous. But converse is not true.

(*iii*) A differentiable function is always continuous. But converse is not true

Necessary Condition (C-R Eq.)

THE NECESSARY CONDITION FOR $f(z)$ TO BE ANALYTIC

Theorem. *The necessary conditions for a function $f(z) = u + iv$ to be analytic at all the points in a region R are*

$$(i)\ \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \qquad (ii)\ \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \text{ provided } \frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial v}{\partial x}, \frac{\partial v}{\partial y} \text{ exist.}$$

These are also known as Cauchy - Riemann equation

Sufficient Condition

SUFFICIENT CONDITION FOR $f(z)$ TO BE ANALYTIC

Theorem. *The sufficient condition for a function $f(z) = u + iv$ to be analytic at all the points in a region R are*

(i) $\dfrac{\partial u}{\partial x} = \dfrac{\partial v}{\partial y}, \qquad \dfrac{\partial u}{\partial y} = -\dfrac{\partial v}{\partial x}$

(ii) $\dfrac{\partial u}{\partial x}, \dfrac{\partial u}{\partial y}, \dfrac{\partial v}{\partial x}, \dfrac{\partial v}{\partial y}$ *are continuous functions of x and y in region R.*

Remember:

1. If a function is analytic in a domain D, then u, v satisfy $C - R$ conditions at all points in D.
2. $C - R$ conditions are necessary but not sufficient for analytic function.
3. $C - R$ conditions are sufficient if the partial derivative are continuous.

Example 4

Example 4. *Determine whether $\frac{1}{z}$ is analytic or not ?* (*R.G.P.V. Bhopal, III Sem., June 2003*)

Solution. Let $w = f(z) = u + iv = \frac{1}{z} \quad \Rightarrow \quad u + iv = \frac{1}{x + iy} = \frac{x - iy}{x^2 + y^2}$

Equating real and imaginary parts, we get

$$u = \frac{x}{x^2 + y^2}, \qquad v = \frac{-y}{x^2 + y^2}$$

$$\frac{\partial u}{\partial x} = \frac{(x^2 + y^2).1 - x.2x}{(x^2 + y^2)^2} = \frac{y^2 - x^2}{(x^2 + y^2)^2}, \qquad \frac{\partial u}{\partial y} = \frac{-2xy}{(x^2 + y^2)^2}.$$

$$\frac{\partial v}{\partial x} = \frac{2xy}{(x^2 + y^2)^2}, \qquad \frac{\partial v}{\partial y} = \frac{y^2 - x^2}{(x^2 + y^2)^2}$$

Thus, $$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \text{ and } \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}.$$

Thus C – R equations are satisfied. Also partial derivatives are continuous except at (0, 0).

Therefore $\frac{1}{z}$ is analytic everywhere except at $z = 0$.

Also $$\frac{dw}{dz} = -\frac{1}{z^2}$$

This again shows that $\frac{dw}{dz}$ exists everywhere except at $z = 0$. Hence $\frac{1}{z}$ is analytic everywhere except at $z = 0$. **Ans.**

Example 5

Example 5. *Show that the function e^x (cos y + i sin y) is an analytic function, find its derivative.* *(R.G.P.V., Bhopal, III Semester, June 2008)*

Solution. Let $e^x(\cos y + i\sin y) = u + iv$

So, $e^x \cos y = u$ and $e^x \sin y = v$ then $\frac{\partial u}{\partial x} = e^x \cos y$, $\frac{\partial v}{\partial y} = e^x \cos y$

$$\frac{\partial u}{\partial y} = -e^x \sin y, \qquad \frac{\partial v}{\partial x} = e^x \sin y$$

Here we see that $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$, $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$

These are $C - R$ equations and are satisfied and the partial derivatives are continuous.

Hence, $e^x(\cos y + i\sin y)$ is analytic.

$$f(z) = u + iv = e^x(\cos y + i\sin y) \text{ and } \frac{\partial u}{\partial x} = e^x \cos y, \quad \frac{\partial v}{\partial x} = e^x \sin y$$

$$f'(z) = \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x} = e^x \cos y + ie^x \sin y = e^x(\cos y + i\sin y) = e^x . e^{iy} = e^{x+iy} = e^z .$$

Which is the required derivative. **Ans.**

C-R Eq. in Polar Form

C–R EQUATIONS IN POLAR FORM

$$\frac{\partial u}{\partial r} = \frac{1}{r}\frac{\partial v}{\partial \theta}$$

$$\frac{\partial u}{\partial \theta} = -r\frac{\partial v}{\partial r}$$

Example

Example 14. *If n is real, show that r^n (cos $n\theta$ + i sin $n\theta$) is analytic except possible when $r = 0$ and that its derivative is*

$$nr^{n-1}\,[\cos(n-1)\,\theta + i\sin(n-1)\,\theta].$$

Solution. Let $w = f(z) = u + iv = r^n(\cos n\theta + i\sin n\theta)$

Here, $u = r^n\cos n\theta, \qquad v = r^n\sin n\theta$

then,
$$\frac{\partial u}{\partial r} = nr^{n-1}\cos n\theta, \qquad \frac{\partial v}{\partial r} = nr^{n-1}\sin n\theta$$
$$\frac{\partial u}{\partial \theta} = -\,nr^n\sin n\theta, \qquad \frac{\partial v}{\partial \theta} = nr^n\cos n\theta$$

Here,
$$\frac{\partial u}{\partial r} = nr^{n-1}\cos n\theta = \frac{1}{r}(nr^n\cos n\theta)$$
$$\frac{\partial u}{\partial r} = \frac{1}{r}\frac{\partial v}{\partial \theta} \qquad \ldots(1)$$

and
$$\frac{\partial v}{\partial r} = nr^{n-1}\sin n\theta \qquad \frac{\partial v}{\partial r} = -\frac{1}{r}(-\,nr^n\sin n\theta)$$
$$\frac{\partial v}{\partial r} = -\frac{1}{r}\frac{\partial u}{\partial \theta} \qquad \ldots(2)$$

Equations (1) and (2) satisfied C-R equations.

We have,
$$\frac{dw}{dz} = (\cos\theta - i\sin\theta)\frac{\partial w}{\partial r} = (\cos\theta - i\sin\theta)\left(\frac{\partial u}{\partial r} + i\,\frac{\partial v}{\partial r}\right)$$
$$= (\cos\theta - i\sin\theta)\,(nr^{n-1}\cos n\theta + inr^{n-1}\sin n\theta)$$
$$= (\cos\theta - i\sin\theta)\,nr^{n-1}\,(\cos n\theta + i\sin n\theta)$$
$$= nr^{n-1}\,\{(\cos n\theta\cos\theta + \sin n\theta\sin\theta) + i\,(\sin n\theta\cos\theta - \cos n\theta\sin\theta)\}$$
$$= nr^{n-1}\,\{\cos(n-1)\,\theta + i\sin(n-1)\theta\}$$

This exists for all finite values of r including zero, except when $r = 0$ and $n \le 1$. **Proved.**

Exercise

EXERCISE 27.1

Determine which of the following functions are analytic:

1. $x^2 + iy^2$ **Ans.** Analytic at all points $y = x$

2. $2xy + i(x^2 - y^2)$ **Ans.** Not analytic

3. $\dfrac{x - iy}{x^2 + y^2}$ **Ans.** Not analytic

4. $\dfrac{1}{(z-1)(z+1)}$ **Ans.** Analytic at all points, except $z = \pm 1$

5. $\dfrac{x - iy}{x - iy + a}$ **Ans.** Not analytic

6. $\sin x \cosh y + i \cos x \sinh y$ **Ans.** Yes, analytic

7. $xy + iy^2$ **Ans.** Yes, analytic at origin

8. Discuss the analyticity of the function $f(z) = z\bar{z} + \bar{z}^2$ in the complex plane, where $\bar{z}$ is the complex conjugate of z. Also find the points where it is differentiable but not analytic.

Ans. Differentiable only at $z = 0$, No where analytic.

9. Show the function of $\bar{z}$ is not analytic any where.

10. If $\begin{cases} \dfrac{x^2 y (y - ix)}{x^4 + y^2}, & \text{when } z \neq 0 \\ 0, & \text{when } z = 0 \end{cases}$

prove that $\dfrac{f(z) - f(0)}{z} \to 0$, as $z \to 0$, along any radius vector but not as $z \to 0$ in any manner.

(AMIETE, Dec. 2010)

11. If $f(z)$ is an analytic function with constant modulus, show that $f(z)$ is constant. *(AMIETE, Dec. 2009)*

Harmonic Function

HARMONIC FUNCTION

(U.P., III Semester 2009-2010)

Any function which satisfies the Laplace's equation is known as a harmonic function.

Theorem. If $f(z) = u + iv$ is an analytic function, then u and v are both harmonic functions.

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \qquad \left(\frac{\partial^2 v}{\partial x \partial y} = \frac{\partial^2 v}{\partial y \partial x}\right)$$

Similarly $$\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0$$

Therefore both u and v are harmonic functions.

Such functions u, v are called **Conjugate harmonic functions if $u + iv$ is also analytic function.**

Example

Example 15. *Define a harmonic function and conjugate harmonic function. Find the harmonic conjugate function of the function U (x, y) = 2x (1 – y). (U.P., III Semester Dec. 2009)*

Solution. See Art. 27.13.

Here, we have $U(x, y) = 2x(1 - y)$ Let V be the harmonic conjugate of U.

By total differentiation

$$dV = \frac{\partial V}{\partial x} dx + \frac{\partial V}{\partial y} dy$$

$$= -\frac{\partial U}{\partial y} dx + \frac{\partial U}{\partial x} dy$$

$$= -(-2x)\, dx + (2 - 2y)\, dy +$$

$$= 2x\, dx + (2\, dy - 2y\, dy) + C$$

$$V = x^2 + 2y - y^2 + C$$

Hence, the harmonic conjugate of U is $x^2 + 2y - y^2 + C$ **Ans.**

Example

Example 16. *Show that the function* $u = \frac{1}{2}\log(x^2+y^2)$ *is harmonic. Find its harmonic conjugate.*

Solution. $u = \frac{1}{2}\log(x^2+y^2)$

$$\frac{\partial u}{\partial x} = \frac{1}{2}\,\frac{1}{x^2+y^2}\,.\,(2x) = \frac{x}{x^2+y^2}\,, \qquad \frac{\partial u}{\partial y} = \frac{y}{x^2+y^2}$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{(x^2+y^2)\,.\,1 - x(2x)}{(x^2+y^2)^2} = \frac{y^2-x^2}{(x^2+y^2)^2}$$

$$\frac{\partial^2 u}{\partial y^2} = \frac{(x^2+y^2)\,.\,1 - y(2y)}{(x^2+y^2)^2} = \frac{x^2-y^2}{(x^2+y^2)^2} \qquad \Rightarrow \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

Hence u is a harmonic function.
Let v be the harmonic conjugate of u.

$$dv = \frac{\partial v}{\partial x}dx + \frac{\partial v}{\partial y}dy = -\frac{\partial u}{\partial y}dx + \frac{\partial u}{\partial x}dy \qquad \text{(By C – R equations)}$$

$$dv = -\frac{y}{x^2+y^2}dx + \frac{x}{x^2+y^2}dy$$

$$dv = \frac{xdy - ydx}{x^2+y^2} = d\left(\tan^{-1}\frac{y}{x}\right)$$

Integrating, we get $v = \tan^{-1}\frac{y}{x} + C$, where C is a real constant.

This is the required harmonic conjugate. **Ans.**

Method for conjugate function

6 METHOD TO FIND THE CONJUGATE FUNCTION

Case I. Given. If $f(z) = u + iv$, and u is known.

To find. v, conjugate function.

Method. We know that $$dv = \frac{\partial v}{\partial x} \cdot dx + \frac{\partial v}{\partial y} \cdot dy \qquad \ldots (1)$$

Replacing $\frac{\partial v}{\partial x}$ by $-\frac{\partial u}{\partial y}$ and $\frac{\partial v}{\partial y}$ by $\frac{\partial u}{\partial x}$ in (1), we get [C-R equations]

$$dv = -\frac{\partial u}{\partial y} \cdot dx + \frac{\partial u}{\partial x} \cdot dy$$

$$v = -\int \frac{\partial u}{\partial y} dx + \int \frac{\partial u}{\partial x} \cdot dy$$

$$\Rightarrow \qquad v = \int M\, dx + \int N\, dy \qquad \ldots (2)$$

where $$M = -\frac{\partial u}{\partial y} \text{ and } N = \frac{\partial u}{\partial x}$$

so that $$\frac{\partial M}{\partial y} = \frac{\partial}{\partial y}\left(-\frac{\partial u}{\partial y}\right) = -\frac{\partial^2 u}{\partial y^2} \text{ and } \frac{\partial N}{\partial x} = \frac{\partial}{\partial x}\left(\frac{\partial u}{\partial x}\right) = \frac{\partial^2 u}{\partial x^2}$$

since u is a conjugate function, so $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$

$$\Rightarrow \qquad -\frac{\partial^2 u}{\partial y^2} = \frac{\partial^2 u}{\partial x^2} \qquad \Rightarrow \qquad \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \qquad \ldots (3)$$

Equation (3) satisfies the condition of an exact differential equation.

Contd…

So equation (2) can be integrated and thus v is determined.

Case II. Similarly, if $v = v(x, y)$ is given

To find out u.

We know that $$du = \frac{\partial u}{\partial x} dx + i \frac{\partial u}{\partial y} dy \qquad \text{... (4)}$$

On substituting the values of $\frac{\partial u}{\partial x}$ and $\frac{\partial u}{\partial y}$ in (4) , we get

$$du = \frac{\partial v}{\partial y} dx - \frac{\partial v}{\partial x} dy$$

On integrating, we get

$$u = \int \frac{\partial v}{\partial y} dx - \int \frac{\partial v}{\partial x} dy \qquad \text{...(5)}$$

(since v is already known so $\frac{\partial v}{\partial y}, \frac{\partial v}{\partial x}$ on R.H.S. are also known)

Equation (5) is an exact differential equation. On solving (5), u can be determined. Consequently $f(z) = u + iv$ can also be determined.

Milne-Thomson Method

Method: 3 [Milne – Thomson method]

(i) $\boxed{\text{To find } f(z) \text{ when } u \text{ is given}}$

$$\text{Let } f(z) = u + iv$$

$$f'(z) = u_x + iv_x$$

$$= u_x - iv_y \text{ [by C—R condition]}$$

$$\therefore f(z) = \int u_x(z,0)dz - i\int u_y(z,0)dz + C \text{ [by Milne—Thomson rule],}$$

Where, C is a complex constant.

(ii) $\boxed{\text{To find } f(z) \text{ when v is given}}$

$$\text{Let } f(z) = u + iv$$

$$f'(z) = u_x + iv_x$$

$$= v_y + iv_x \quad \text{[by C—R condition]}$$

$$\therefore f(z) = \int v_y(z,0)dz + i\int v_x(z,0)dz + C \text{ [by Milne—Thomson rule],}$$

Where, C is a complex constant.

Example

Example 20. *Let $f(z) = u(x, y) + iv(x, y)$ be an analytic function. If $u = 3x - 2xy$, then find v and express $f(z)$ in terms of z.*

Solution. Here, we have $u = 3x - 2xy$

$$\frac{\partial u}{\partial x} = 3 - 2y, \qquad \frac{\partial u}{\partial y} = -2x$$

We know that
$$dv = \frac{\partial v}{\partial x}dx + \frac{\partial v}{\partial y}dy \qquad \text{(Total differentiation)}$$

$$= \left(-\frac{\partial u}{\partial y}\right)dx + \left(\frac{\partial u}{\partial x}\right)dy \qquad \left(\frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}, \ \frac{\partial v}{\partial y} = \frac{\partial u}{\partial x}\right)$$

$$= 2x\,dx + (3 - 2y)\,dy$$

$$v = \int 2x\,dx + \int (3-2y)\,dy = x^2 + 3y - y^2 + c$$

$$f(z) = u(x, y) + iv(x, y) = (3x - 2xy) + i(x^2 + 3y - y^2 + c)$$

$$= (ix^2 - iy^2 - 2xy) + (3x + 3yi) + ic = i(x^2 - y^2 + 2ixy) + 3(x + iy) + ic$$

$$= i(x + iy)^2 + 3(x + iy) + ic = iz^2 + 3z + ic$$ **Ans.**

Which is the required expression of $f(z)$ in terms of z.

Example

Example: 3.22 Construct the analytic function $f(z)$ for which the real part is $e^x \cos y$.

Solution:

Given $u = e^x \cos y$

$\Rightarrow u_x = e^x \cos y$ $[\because \cos 0 = 1]$

$\Rightarrow u_x(z,0) = e^x$

$\Rightarrow u_y = e^x \cos y$ $[\because \sin 0 = 0]$

$\Rightarrow u_y(z,0) = 0$

$\therefore f(z) = \int u_x(z,0)dz - i\int u_y(z,0)dz + C$ [by Milne–Thomson rule],

Where, C is a complex constant.

$\therefore f(z) = \int e^z\, dz - i\int 0\, dz + C$

$= e^z + C$

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Example

Example: 3.24 Determine the analytic function where real part is

$u = x^3 - 3xy^2 + 3x^2 - 3y^2 + 1.$ **[Anna, May 2001]**

Solution:

$$\text{Given} \quad u = x^3 - 3xy^2 + 3x^2 - 3y^2 + 1$$

$$u_x = 3x^2 - 3y^2 + 6x$$

$$\Rightarrow u_x(z,0) = 3z^2 - 0 + 6z$$

$$u_y = 0 - 6xy + 0 - 6y$$

$$\Rightarrow u_y(z,0) = 0$$

$$f(z) = \int u_x(z,0)dz - i\int u_y(z,0)dz + C \quad \text{[by Milne−Thomson rule],}$$

Where, C is a complex constant.

$$f(z) = \int (3z^2 + 6z)dz - i\int 0 + dz + C$$

$$= 3\frac{z^2}{3} + 6\frac{z^2}{2} + C$$

$$= z^3 + 3z^2 + C$$

Exercise

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EXERCISE 27.2

Show that the following functions are harmonic and determine the conjugate functions.

1. $u = 2x(1-y)$ **Ans.** $v = x^2 - y^2 + 2y + C$ **2.** $u = 2x - x^3 + 3xy + 3xy^2$

Ans. $v = -\frac{3}{2}(x^2 - y^2) + \frac{3}{2}y^2 + y^3 + 2y + C$

Determine the analytic function, whose real part is

3. $\log\sqrt{x^2+y^2}$ **Ans.** $\log z + C$ **4.** $\cos x \cosh y$ **Ans.** $\cos z + c$

5. $e^{-x}(\cos y + \sin y)$ *(AMIETE, June 2010)* **Ans.** $e^{-z}(1 + i + C$

6. $e^{2x}(x\cos 2y - y\sin 2y)$ **Ans.** $ze^{2z} + iC$ **7.** $e^{-x}(x\cos y + y\sin y)$ and $f(0) = i$. **Ans.** $ze^{-z} + i$

Determine the analytic function, whose imaginary part is

8. $v = \log(x^2+y^2) + x - 2y$ *(GBTU, III Sem. April 2012)* **Ans.** $2i\log z - (2-i)z + C$

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Elementary Analytic Functions

Elementary analytic functions are functions that are analytic in their domains and frequently used in complex analysis. The most common elementary analytic functions include:

- Exponential Function
- Trigonometric Functions
- Logarithmic Function

These functions possess derivatives at every point in their domains and play an important role in mathematical analysis and engineering applications.

Exponential Function

The exponential function is defined as:

$$f(z) = e^z$$

Properties:

- Entire function (analytic everywhere in the complex plane).
- Derivative:

$$\frac{d}{dz}(e^z) = e^z$$

- Periodic in the imaginary direction:

$$e^{z+2\pi i} = e^z$$

- Non–zero for all complex numbers.

Trigonometric Functions

The complex trigonometric functions are defined using exponential functions:

$$\sin z = \frac{e^{iz} - e^{-iz}}{2i}$$

$$\cos z = \frac{e^{iz} + e^{-iz}}{2}$$

Properties:

- Both $\sin z$ and $\cos z$ are entire functions.
- Periodic functions with period 2π.
- Derivatives:

$$\frac{d}{dz}(\sin z) = \cos z$$

$$\frac{d}{dz}(\cos z) = -\sin z$$

Logarithmic Function

The complex logarithmic function is defined as:

$$\log z = \ln |z| + i \arg(z)$$

Properties:

- Multi–valued function in the complex plane.
- Analytic except at $z = 0$.
- Derivative:

$$\frac{d}{dz}(\log z) = \frac{1}{z}$$

- Principal value:

$$\text{Log } z = \ln |z| + i(z)$$

Properties

- Exponential, trigonometric, and logarithmic functions are fundamental analytic functions.
- Exponential and trigonometric functions are entire.
- Logarithmic function is analytic except at the origin and involves branch cuts.

28.1 GEOMETRICAL REPRESENTATION

To draw a curve of complex variable (x, y) on z-plane we take two axes *i.e.*, one real axis and the other imaginary axis. A number of points (x, y) are plotted on z-plane, by taking different value of z (different values of x and y). The curve C is drawn by joining the plotted points. The diagram obtained is called *Argand diagram* in z-plane.

But a complex function $w = f(z)$ *i.e.*, $(u + iv) = f(x + iy)$ involves four variables x, y and u, v.

A figure of only three dimensions (x, y, z) is possible in a plane. A figure of four dimensional region for x, y, u, v is not possible.

So, we choose two complex planes z-plane and w-plane. In the w-plane we plot the corresponding points $w = u + iv$. By joining these points we have a corresponding curve C' in w-plane.

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28.2 TRANSFORMATION

For every point (x, y) in the z-plane, the relation $w = f(z)$ defines a corresponding point (u, v) in the w-plane. We call this "transformation or mapping of z-plane into w-plane". If a point z_0 maps into the point w_0, w_0 is also known as the image of z_0.

If the point $P(x, y)$ moves along a curve C in z-plane, the point $P'(u, v)$ will move along a corresponding curve C' in w-plane, then we say that a curve C in the z-plane is mapped into the corresponding curve C' in the w-plane by the relation $w = f(z)$.

Example

Example 1. *Transform the rectangular region ABCD in z-plane bounded by $x = 1$, $x = 3$; $y = 0$ and $y = 3$. Under the transformation $w = z + (2 + i)$.*

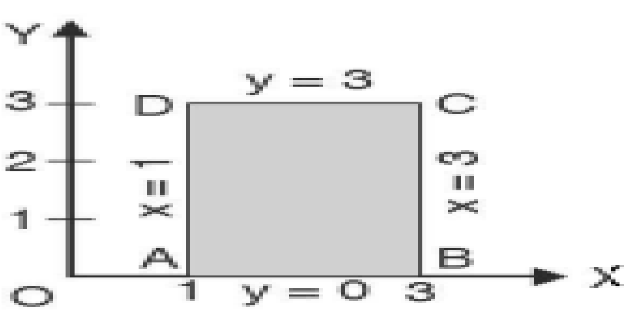
Solution. Here, $$w = z + (2 + i)$$

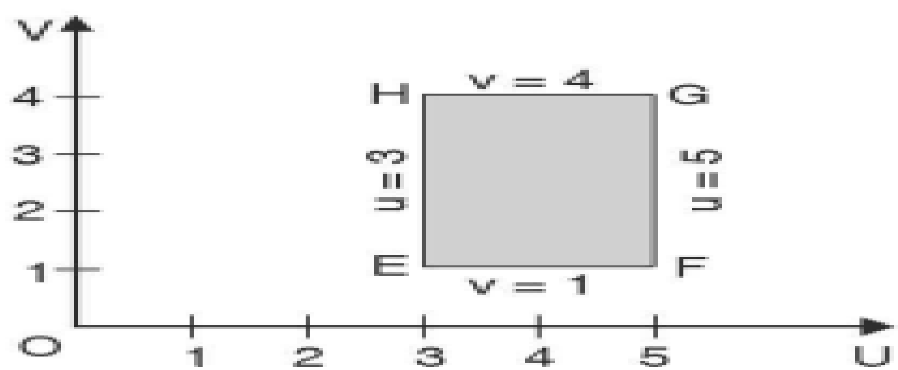
$$\Rightarrow \quad u + iv = x + iy + (2 + i)$$
$$= (x + 2) + i(y + 1)$$

By equating real and imaginary quantities, we have $u = x + 2$ and $v = y + 1$.

z-plane	w-plane	z-plane	w-plane
x	$u = x + 2$	y	$v = y + 1$
1	$= 1 + 2 = 3$	0	$= 0 + 1 = 1$
3	$= 3 + 2 = 5$	3	$= 3 + 1 = 4$

Here the lines $x = 1$, $x = 3$; $y = 0$ and $y = 1$ in the z-plane are transformed onto the line $u = 3$, $u = 5$; $v = 1$ and $v = 4$ in the w-plane. The region $ABCD$ in z-plane is transformed into the region $EFGH$ in w-plane.





Conformal Transformation

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28.3 CONFORMAL TRANSFORMATION (*U.P. III Semester Dec., 2006, 2005*)

Let two curves C, C_1 in the z-plane intersect at the point P and the corresponding curve C', C_1' in the w-plane intersect at P'. **If the angle of intersection of the curves at P in z-plane**

is the same as the angle of intersection of the curves of w-plane at P' in magnitude and sense, then the transformation is called conformal:

conditions: (*i*) $f(z)$ is analytic. (*ii*) $f'(z) \neq 0$ Or

If the sense of the rotation as well as the magnitude of the angle is preserved, the transformation is said to be **conformal**.

If only the magnitude of the angle is preserved, transformation is **Isogonal**.

28.4 THEOREM. If *f*(*z*) is analytic, mapping is conformal (*U.P. III Semester Dec. 2005*)

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Example

Example 3. *If* $u = 2x^2 + y^2$ *and* $v = \dfrac{y^2}{x}$, *show that the curves u = constant and v= constant cut orthogonally at all intersections but that the transformation w = u + iv is not conformal.*

(Q. Bank U.P. III Semester 2002)

Solution. For the curve, $2x^2 + y^2 = u$

$$2x^2 + y^2 = \text{constant} = k_1 \text{ (say)} \qquad \ldots(1)$$

Differentiating (1), we get

$$4x + 2y\frac{dy}{dx} = 0 \quad \Rightarrow \quad \frac{dy}{dx} = -\frac{2x}{y} \qquad \ldots(2)$$

$$\frac{y^2}{x} = v$$

For the curve, $\dfrac{y^2}{x} = \text{constant} = k_2$ (say),

$$\Rightarrow \quad y^2 = k_2 x. \qquad \ldots(3)$$

Differentiating (3), we get

$$2y\frac{dy}{dx} = k_2 \quad \Rightarrow \quad \frac{dy}{dx} = \frac{k_2}{2y} = \frac{y^2}{x} \times \frac{1}{2y} = \frac{y}{2x} \qquad \ldots(4)$$

From (2) and (4), we see that

$$m_1 m_2 = \left(\frac{-2x}{y}\right)\left(\frac{y}{2x}\right) = -1$$

Hence, two curves cut orthogonally.

However, since

$$\frac{\partial u}{\partial x} = 4x, \qquad \frac{\partial u}{\partial y} = 2y$$

$$\frac{\partial v}{\partial x} = -\frac{y^2}{x^2}, \qquad \frac{\partial v}{\partial y} = \frac{2y}{x}$$

The Cauchy-Riemann equations are not satisfied by u and v.

Hence, the function $u + iv$ is not analytic. So, the transformation is not conformal. **Proved**

Example

Example: 3.37 Find the image of the circle $|z| = 1$ by the transformation $w = z + 2 + 4i$

Solution:

Given $w = z + 2 + 4i$

$$(i.e.)\ u + iv = x + iy + 2 + 4i$$

$$= (x + 2) + i(y + 4)$$

Equating the real and imaginary parts, we get

$$u = x + 2,\ v = y + 4,$$

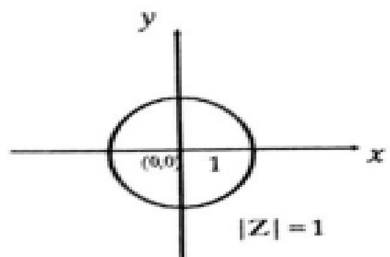
$$x = u - 2,\ y = v - 4,$$

Given $|z| = 1$

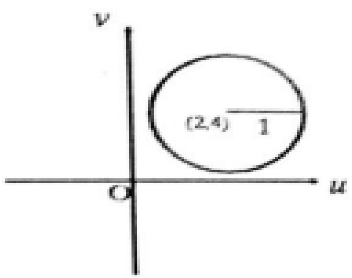
$$(i.e.)\ x^2 + y^2 = 1$$

$$(u - 2)^2 + (v - 4)^2 = 1$$

Hence, the circle $x^2 + y^2 = 1$is mapped into $(u - 2)^2 + (v - 4)^2 = 1$ in w plane which is also a circle with centre (2, 4)and radius 1.



z-Plane



w-Plane

Example

Example: 3.39 Find the image of the circle $|z| = \lambda$ under the transformation $w = 5z$.

Solution:

Given $w = 5z$

$$|w| = 5|z|$$

i.e., $|w| = 5\lambda \quad [\because |z| = \lambda]$

Hence, the image of $|z| = \lambda$ in the z plane is transformed into $|w| = 5\lambda$ in the w plane under the transformation $w = 5z$.

Mobius transformations and their properties

The transformation $w = \frac{az+b}{cz+d}, ad - bc \neq 0$ where a, b, c, d are complex numbers, is called a bilinear transformation.

This transformation was first introduced by A.F. Mobius, So it is also called Mobius transformation.

A bilinear transformation is also called a linear fractional transformation because $\frac{az+b}{cz+d}$ is a fraction formed by the linear fuctions $az - b$ and $cz + d$.

Theorem: 1 Under a bilinear transformation no two points in z plane go to the same point in w plane.

Theorem: 2 The bilinear transformation which transforms $z_1, z_2, z_3,$ into w_1, w_2, w_3 is

$$\frac{(w-w_1)(w_2-w_3)}{(w-w_3)(w_2-w_1)} = \frac{(z-z_1)(z_2-z_3)}{(z-z_3)(z_2-z_1)}$$

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The point $z = \frac{-d}{c}$ corresponds to the point at infinity in the w plane.

Note: (5) The cross ratio of four points

$$\frac{(w_1 - w_2)(w_3 - w_4)}{(w_2 - w_3)(w_4 - w_1)} = \frac{(z_1 - z_2)(z_3 - z_4)}{(z_2 - z_3)(z_4 - z_1)}$$ is invariant under bilinear transformation.

Note: (6) If one of the points is the point at infinity the quotient of those difference which involve this points is replaced by 1.

Suppose $z_1 = \infty$, then we replace $\frac{z-z_1}{z_2-z_1}$ by 1 (or)Omit the factors involving ∞

Example

Example: 3.59 Find the fixed points of $w = \frac{2zi+5}{z-4i}$.

Solution:

The fixed points are given by replacing *w by z*

$$z = \frac{2zi+5}{z-4i}$$

$$z^2 - 4iz = 2zi + 5\ ;\ z^2 - 6iz - 5 = 0$$

$$z = \frac{6i \pm \sqrt{-36+20}}{2} \qquad \therefore z = 5i, i$$

Example

Example: 3.60 Find the invariant points of $w = \frac{1+z}{1-z}$

Solution:

The invariant points are given by replacing *w by z*

$$z = \frac{1+z}{1-z}$$

$$\Rightarrow z - z^2 = 1 + z$$

$$\Rightarrow z^2 = -1$$

$$\Rightarrow z = \pm i$$

Example

Example: 3.61 Obtain the invariant points of the transformation $w = 2 - \frac{2}{z}$. [Anna, May 1996]

Solution:

The invariant points are given by

$$z = 2 - \frac{2}{z}; \qquad z = \frac{2z-2}{z}$$

$$z^2 = 2z - 2; \qquad z^2 - 2z + 2 = 0$$

$$z = \frac{2 \pm \sqrt{4-8}}{2} = \frac{2 \pm \sqrt{-4}}{2} = \frac{2 \pm 2i}{2} = 1 \pm i$$

Thank You

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